Chapter 5 Absolute System

NOTICE

- The absolute encoder must carry a battery box.

5.1 Absolute System Setting

Overview

The absolute encoder, which carries a resolution of 131072 (2^{17}) PPR, detects the motor position within one revolution and counts the number of revolutions, with 16-bit multi-turn data saved.

The absolute encoder system works in the position control, speed control, and torque control modes. When the servo drive is powered off, the encoder battery serves as the power supply to enable the encoder to back up data. The servo drive therefore can calculate the absolute mechanical position through the encoder after power-on, removing the need for homing.

When A6-EC series servo drive matches the absolute encoder, set 2000.08h (C00.07) to (absolute system selection) based on actual conditions. Er208 (encoder battery fault) will be reported when the battery is connected for the first time. In this case, set 2031.11h (F31.10) to reset the encoder fault, and then perform homing.

NOTICE

- When the value of 2000.02h (C00.01) (rotation direction) or 2031.11h (F31.10) (absolute encoder reset selection), or the electronic gear ratio is changed, the mechanical position will change abruptly, requiring a homing operation.
- After homing is done, the deviation between the mechanical absolute position and that saved in the encoder will be calculated automatically and saved in the EEPROM of the servo drive.

Absolute system setting

Set 2000.08h (C00.07) to select the absolute position mode.

Related parameters:

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
08h	C00.07	Absolute mode	0: Incremental position mode 1: Absolute position linear mode 2: Absolute position linear infinite mode 3: Absolute position single-turn mode 4: Absolute position rotation mode 5: Absolute mechanical single-turn mode (operating direction selectable)	0 to 5	0	-	U16	At stop	Upon re-power-on

Encoder feedback data

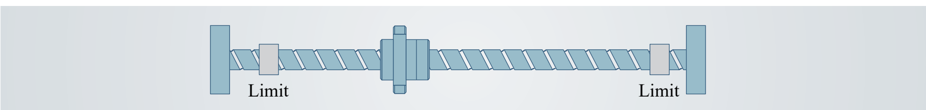
The feedback data of an absolute encoder includes the number of revolutions and the motor position within one revolution. In incremental position mode, the number of revolutions will not be counted.

Related parameters:

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
1Dh	U40.1C	Encoder single-turn data	-	-2^{31} to $(2^{31}-1)$	0	P	I32	-	Immediately
1Fh	U40.1E	Encoder multi-turn position data	-	0 to 65535	0	Rev	U16	-	Immediately
21h	U40.20	Encoder multi-turn data (low 32 bits)	-	-2^{31} to $(2^{31}-1)$	0	P	I32	-	Immediately
23h	U40.22	Encoder multi-turn data (high 32 bits)	-	-2^{31} to $(2^{31}-1)$	0	P	I32	-	Immediately

5.2 Absolute Position Linear Mode

This mode applies to scenarios where the axis travel range is fixed without multi-turn data overflow.



The value range of 2000.08h=1 (C00.07) (encoder multi-turn data) in absolute position linear infinite mode is 0 to 65535. If the number of forward or reverse revolutions is greater than 65535, ErA01 (encoder multi-turn counting overflow) will occur. In this case, set 2031.11h (F31.10) to 4 to reset the multi-turn data, and then perform homing again. In special occasions, you can set 2000.08h (C00.07) to 2 in absolute position linear infinite mode to prevent the encoder overflow alarm.

Related parameters:

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
17h	U40.16	Absolute position feedback (reference unit)	-	-2^{31} to $(2^{31}-1)$	0	Unit in application	I32	-	Immediately
25h	U40.24	Absolute position feedback (encoder unit) (low 32 bits)	-	-2^{31} to $(2^{31}-1)$	0	P	I32	-	Immediately
27h	U40.26	Absolute position feedback (encoder unit) (high 32 bits)	-	-2^{31} to $(2^{31}-1)$	0	P	I32	-	Immediately

Index	Sub-index	Name	Access	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
6063h	0	Position feedback	RO	TPDO	I32	Encoder unit	-	-	-	-
6064h	0	Position feedback	RO	TPDO	I32	Reference unit	-	-	-	-

5.3 Absolute Position Rotation Mode

This mode applies in cases where the load travel range is unlimited and the number of unidirectional revolutions is less than 32767, as shown in the following figure.

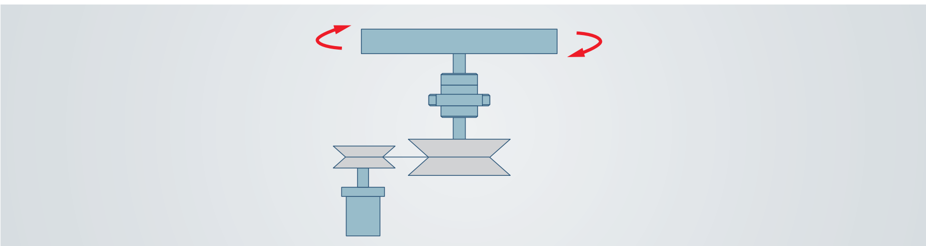
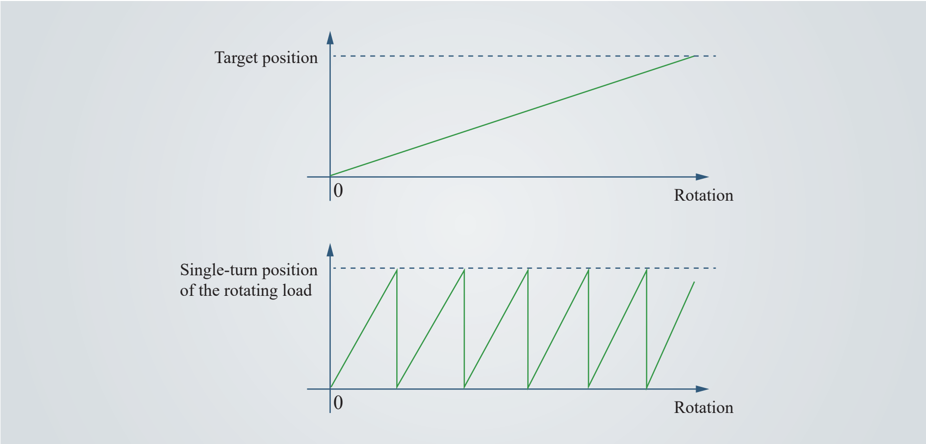
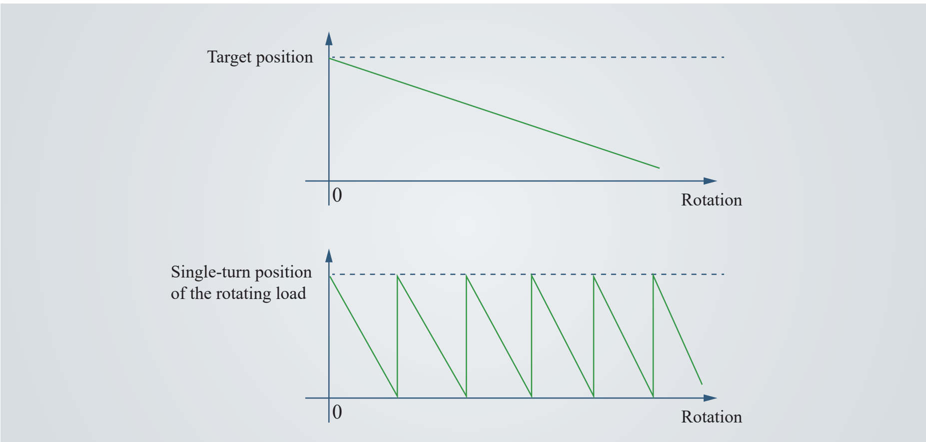


Figure 5-1 Rotating load

The single-turn position range of the rotating load is 0 to (R_M - 1) (R_M: Encoder pulses per load revolution). When the gear ratio is 1:1, the variation law of the target position and the single-turn position of the rotating load during forward operation is as follows.



The variation law of the target position and the single-turn position of the rotating load during reverse operation is as follows.



When the motor works in the absolute position rotation mode while the servo drive works in HM mode, the home offset range is 0 to (R_M - 1).

NOTICE

- The servo drive calculates the upper limit of the mechanical absolute position using 2010.1Bh (C10.1A) and 2010.1Dh (C10.1C) first. If 2010.1Bh (C10.1A) and 2010.1Dh (C10.1C) are both set to 0, the servo drive employs the electronic gear ratio 2010.19h (C10.18) and 2010.1Ah (C10.19) for calculation.

Related parameters:

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
19h	C10.18	Numerator of electronic gear ratio in rotation mode	-	1 to 65535	1	-	U16	At stop	Immediately
1Ah	C10.19	Denominator of electronic gear ratio in rotation mode	-	1 to 65535	1	-	U16	At stop	Immediately
1Bh	C10.1A	Upper limit of mechanical absolute position in rotation mode (low 32 bits)	-	0 to (2^{32}-1)	0	P	U32	At stop	Immediately
1Dh	C10.1C	Upper limit of mechanical absolute position in rotation mode (high 32 bits)	-	0 to (2^{32}-1)	0	P	U32	At stop	Immediately
29h	U40.28	Position feedback in rotation mode (reference unit) (low 32 bits)	-	-2^{31} to (2^{31}-1)	0	Unit in application	I32	-	Immediately
2Bh	U40.2A	Position feedback in rotation mode (encoder unit) (low 32 bits)	-	-2^{31} to (2^{31}-1)	0	P	I32	-	Immediately
2Dh	U40.2C	Position feedback in rotation mode (encoder unit) (high 32 bits)	-	-2^{31} to (2^{31}-1)	0	P	I32	-	Immediately

Index	Sub-index	Name	Access	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
6063h	0	Position feedback	RO	TPDO	I32	Encoder unit	-	-	-	-
6064h	0	Position feedback	RO	TPDO	I32	Reference unit	-	-	-	-

5.4 Absolute Position Single-turn Mode

In this mode, the absolute encoder needs no battery and does not record the number of motor revolutions. The input range for the target position of the EtherCAT host controller is -2^{31} to (2^{31} - 1), with no need for modulus operation on the encoder resolution. The initial position for feedback upon each power-on is the single-turn absolute position. For a 17-bit encoder, the value range of 6064 is 0 to (2^{17} - 1) for each power-on. If the value of C00.07 remains 3 (absolute position single-turn mode), after homing (homing mode of the servo drive, that is, 6060 = 6), the servo drive saves the absolute position of the encoder after homing and the value of 607C (home offset). Upon power-up again, the value of 6064 (current position feedback) will be calculated based on the coordinate system after the last homing operation, eliminating the need for re-homing.

If 60E6h (absolute position single-turn mode) is inactive, after successful homing, the value of 6064 (position feedback) is equal to that of 607C (home offset).

Related parameters:

Index	Sub-index	Name	Access	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
6063h	0	Position feedback	RO	TPDO	I32	Encoder unit	-	-	-	-
6064h	0	Position feedback	RO	TPDO	I32	Reference unit	-	-	-	-

5.5 Precautions for Use of the Absolute system Battery Box

Er208 (encoder battery fault) will be reported when the battery is connected for the first time. In this case, set 2031.11h (F31.10) to reset the encoder fault, power on again, and then perform the absolute position device operation.

When the battery voltage detected is lower than 3.0 V, ALF90 (encoder battery warning) will be reported. In this case, replace the battery based on the following steps:

① Power on the servo drive and make it stay in the non-operational state.
② Replace the battery.
③ After ALF90 (encoder battery warning) is automatically cleared, if no other warning/fault occurs, you can continue operating the servo drive.

If you replace the battery after power-off, Er208 (encoder battery fault) will be reported, with the multi-turn data changed abruptly. In this case, set 2031.11h (F31.10) to 4 to reset the fault, and then perform homing again.

Ensure the motor speed does not exceed 6000 RPM after the servo drive is powered off. This is to enable the encoder to save the position data accurately.

Keep the battery in environments within the required ambient temperature and ensure the battery is in reliable contact with sufficient power reserved. Failure to comply may result in encoder data loss.

NOTICE

- The absolute position saved by the encoder changes abruptly after multi-turn data reset. In this case, perform mechanical homing.

Related parameters:

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
11h	F31.10	Encoder data reset	-	0 to 31	0	-	U16	At stop	Immediately